



# Evaluating expressions with and without changing the subject

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1 Substitute means to put in place of another. In algebra, substitution can be used to replace \_\_\_\_\_ with values. Tick 1 correct answer

☐ constants

☐ equations

☐ solutions

☐ variables

2  $A = \frac{1}{2}(a + b)h$  is the formula for area of a trapezium. It can be rearranged to

$\frac{2A}{h} - a = b$  in order to make b the \_\_\_\_\_ of the formula. Fill in the blank

3 The formula for area of a triangle is  $A = \frac{1}{2}bh$ . The area of a triangle with base 7.5 cm and height 6 cm is \_\_\_\_\_  $\text{cm}^2$  Fill in the blank

4 The formula  $s = \frac{d}{t}$  can be rearranged to  $t = \frac{d}{s}$  or  $d = s \times t$ . If  $d = 12$  and  $s = 3$ , find t.

Tick 1 correct answer

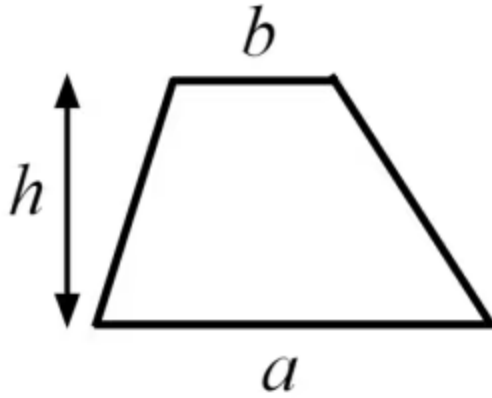
☐ 36

☐ 4

☐  $\frac{1}{4}$



- 5 Find the area of a trapezium with  $h = 7$ ,  $a = 3.2$ ,  $b = 4.8$ . Tick **1** correct answer



$$A = \frac{1}{2}(a + b)h$$

☐  $A = \frac{1}{2}(3.2 + 4.8) + 7 = 11$  square units

☐  $A = \frac{1}{2}(3.2 + 4.8) \times 7 = 28$  square units

☐  $A = 2(3.2 + 4.8) \times 7 = 112$  square units

☐  $A = \frac{1}{2}(3.2 \times 4.8) \times 7 = 53.76$  square units

- 6  $A = \frac{1}{2}(a + b)h$  is the formula for area of a trapezium. Select all the correct rearrangements of this formula. Tick **2** correct answers

☐  $\frac{2A}{h} - b = a$

☐  $\frac{Ah}{2} - a = b$

☐  $\frac{A}{2h} - a = b$

☐  $\frac{2h}{A} - a = b$

☐  $\frac{2A}{a + b} = h$

